# ACHYARUL MABRURI

**Semiconductor Device Researcher · Emerging Memory & Neuromorphic Computing · Youth Leader & Mentor**

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## PROFILE

Graduate research student at Korea University's Advanced Semiconductor Laboratory on the SK Global Scholarship. Works across the device-to-system stack: fabricating and characterizing emerging memory devices, modeling them, and testing what they mean for neuromorphic and in-memory computing hardware. Fully funded study and research in India, Russia, Korea and Singapore; active in youth development since 2018, leading an NGO that has mentored 100+ young people. Aiming for a PhD on reliable emerging memory devices for in-memory and neuromorphic computing.

## EDUCATION

**Korea University, Seoul, Republic of Korea** 2025 – present

*M.S. (Research), Semiconductor and Nano Systems, School of Electrical Engineering*

- **SK Global Scholarship** (Korea University / SK hynix), full scholarship. Advanced Semiconductor Laboratory under Prof. Tae Geun Kim.

**Delhi Technological University, New Delhi, India** 2021 – 2025

*B.Tech., Engineering Physics. First Division with Distinction*

- **ICCR Scholarship** (Government of India), full scholarship. Thesis: "Temperature Dependence of Complete Photonic Band Gap in 2D Photonic Crystal Structures."

## RESEARCH EXPERIENCE

**Graduate Researcher, Advanced Semiconductor Laboratory, Korea University** 2025 – present

- Trained in CBRAM and PRAM fabrication; now fabricate and characterize lead-free metal-halide RRAM devices with a teammate (ongoing) by sputtering and Keithley 4200-SCS I–V and pulse measurements; analyze switching, variability and conduction mechanisms.
- Investigate selector concepts for RRAM crossbar arrays to suppress sneak-path currents; simulate devices and arrays in LTspice, Python and COMSOL Multiphysics.

**Research Fellow, InteRussia Engineering Fellowship, Novosibirsk State University, Russia** Jun 2025

- Selected fellow in Modern Quantum and Information Technologies in Electronics and Photonics; trained with an international cohort in quantum mechanics and in simulating quantum effects in electronic and photonic devices.

**Researcher, Advanced Simulation Lab, TIFAC-CORE, Delhi Technological University** 2024 – 2025

- Modeled temperature-dependent photonic band gaps in Si/GaAs photonic crystals for sensing (MPB, MEEP); published in the Journal of Modern Optics.

**Research Intern, Silicon Photonics Research Group, LNMIIT Jaipur, India** May – Jul 2024

- Modeled carrier mobility and the indirect-to-direct band-gap crossover in $Si_{1-x}Sn_x$ alloys in MATLAB.

## KEY RESEARCH PROJECTS

**Metal-halide RRAM: from measured device to crossbar and neural network** 2026

- Fitted conduction and weight-update models to lab CsX RRAM data; simulated passive crossbars with wire resistance and ADCs; a binary CNN kept 91.5% CIFAR-10 accuracy on device states.

**Phase-change heterostructure (PCH) memory for in-memory computing** 2026

- Built a device-to-chip pipeline for published multi-level PCH devices ($NiTe_2/MoTe_2$, $MoS_2$, $HfTe_2$, $TiTe_2$) vs. a GST baseline: crossbar MAC with drift and IR drop, device-aware VGG8 on CIFAR-10, and NeuroSim TOPS/W.

**Drift compensation for phase-change-memory analog inference** 2026

- Simulated one year of PCM drift in VGG8 inference with IBM aihwkit; global drift compensation plus adaptive batch-norm restored accuracy from 53% to 90%.

**NEUROM: memristive reservoir-computing benchmark across six device models** 2025 – 2026

- Echo-state reservoirs with STDP initialization under Monte Carlo device variability; tasks: sequential MNIST, Mackey–Glass forecasting, spoken digits.

**Memristor compact-model framework and reliability study** 2025 – 2026

- Python library of 21 compact models (window-function, Yakopcic, TEAM, VTEAM, Knowm); identified and fixed three hidden failure modes that silently corrupt LTP/LTD simulations.

**Spiking neural network on measured memristor conductance levels** 2026

- Trained a leaky integrate-and-fire network in snnTorch (96.1% on MNIST) and tested it on three real devices' conductance levels.

**Self-rectifying memristor crossbar pipeline** 2026

- Replicated Sharma et al. (Advanced Materials, 2026) and applied it to three measured devices (readout margin, array size, CNN accuracy).

**Physical reservoir computing with Ag/$MoS_2$/Au CBRAM (LTspice + Python)** 2026

- Replicated the Tanaka & Nakane (2022) reservoir; compared discrete-switching CBRAM with a continuous Yakopcic model.

**Machine-learning surrogate for STDP in drift–diffusive memristor synapses** 2026

- Simulated 2,875 circuit configurations; XGBoost models (99.2% accuracy) map STDP behavior without re-simulation.

**Reproductions and tools** 2025 – 2026

- Reproduced Vourkas (2025) netlists NL1–NL5, the Scientific Reports (2026) RRAM compact model and a lead-free NKN/$HfO_2$ STDP model; ported the TaOx (Zhong 2021) and rebuilt the $ZnPS_3$ (Ji 2025) memristor reservoirs, with control runs; β-VAE reproduction as a course project (with J. Oh).
- Extended the open-source PyNeuroMemSim (LTspice–Python) simulator to four memristor models; ported Keithley 4200-SCS scripts from C to Python; reproduced ML prediction of SET voltage in honey-based RRAM.

## LEADERSHIP & PROFESSIONAL EXPERIENCE

**Managing Director, Gerakan Tunas Bangsa Indonesia (youth-development NGO)** Dec 2021 – Aug 2024

- Led and managed the NGO's day-to-day operations, programs and volunteer team; formulated and set its policies and core values.
- Coordinated internal and external stakeholders to keep the movement active, empowering and sustainable.
- Supervised mentoring, coaching and scholarship-awareness activities reaching 100+ young people.

**Human Quality & Development Lead, Gerakan Tunas Bangsa Indonesia** 2018 – 2021

- Standardized member-development quality and created a system to evaluate each mentee's progress.
- Launched new programs (English classes, coaching, sharing sessions, webinars and workshops) for 100+ mentees.

**Speaker & Mentor: Scholarships, Higher Education and Personal Development** 2021 – present

- Invited speaker, e.g. “Unlock Your Dreams Through Education” (2026); “Developing a Successful Mindset Towards a Dream” (2025); ICCR scholarship mentoring for ~100 participants with PPI India.

**Board of Supervisors, Indonesian Students Association in India (PPI India)** Jul 2024 – Jun 2025

- Supervised the association's programs and governance for Indonesian students across India.

**Asia Region Representative, DTU International Students Society** Nov 2022 – Aug 2024

- Represented international students from the Asia region in the society's Chamber of Representatives.

## INTERNATIONAL PROGRAMS & FELLOWSHIPS

- **International Festival of Youth (IFY) 2026**, Yekaterinburg, Russia: Indonesian delegate — Sep 2026
- **SEA–Europe Youth Partnership for the Goals 2026**, Singapore (ASEF, KU Leuven) — 2026
- **Honorary Young Diplomat**, Jeju International Future Generation Forum (UNITAR), Korea — Nov 2025
- **Aspire Leaders Program**, Cohort 1 — 2025
- **Kectil Leadership Program** (Kectil Colleague) and **UPG Leadership Program** (Leadership Candidate) — 2023

## HONORS & AWARDS

- **SK Global Scholarship**, Korea University / SK hynix (full scholarship) — 2025 – present
- **ICCR Scholarship**, Indian Council for Cultural Relations, Government of India (full scholarship) — 2021 – 2025
- **Top 6 Nominee**, Youth Change Maker Award; **Top 27**, Paragon Innovation Awards — 2021
- **Best Candidate**, LeadID (Leaders Academy Indonesia) by pemimpin.id — 2021

## PUBLICATIONS & PRESENTATIONS

- A. Mabruri et al., “Investigation of Temperature Dependence of Complete Photonic Band Gap in 2D Photonic Crystal with Sensing Application,” *Journal of Modern Optics*, 2025. DOI: 10.1080/09500340.2025.2542218
- Book chapter (accepted; from my poster at the 3rd AFMD conference, 2025): temperature dependence of complete photonic band gaps in 2D photonic crystals. Springer Proceedings in Materials, forthcoming Feb 2027.

## SKILLS & CERTIFICATIONS

**Lab & devices:** sputtering, photolithography, thermal evaporator, RRAM fabrication, Keithley 4200-SCS I–V, pulse testing

**Simulation & AI:** Python (PyTorch, snnTorch, scikit-learn), MATLAB, LTspice, COMSOL Multiphysics, MPB/MEEP; neural networks and data analysis

**Certifications:** Quantum Machine Learning, Hasso Plattner Institute (2023); Molecules and Materials for the Energy of Tomorrow, Université Paris-Saclay (2023); QSilver35 Quantum Computing & Programming, QWorld (2026)

**Languages:** Javanese and Indonesian (native), English (fluent), Korean (beginner)